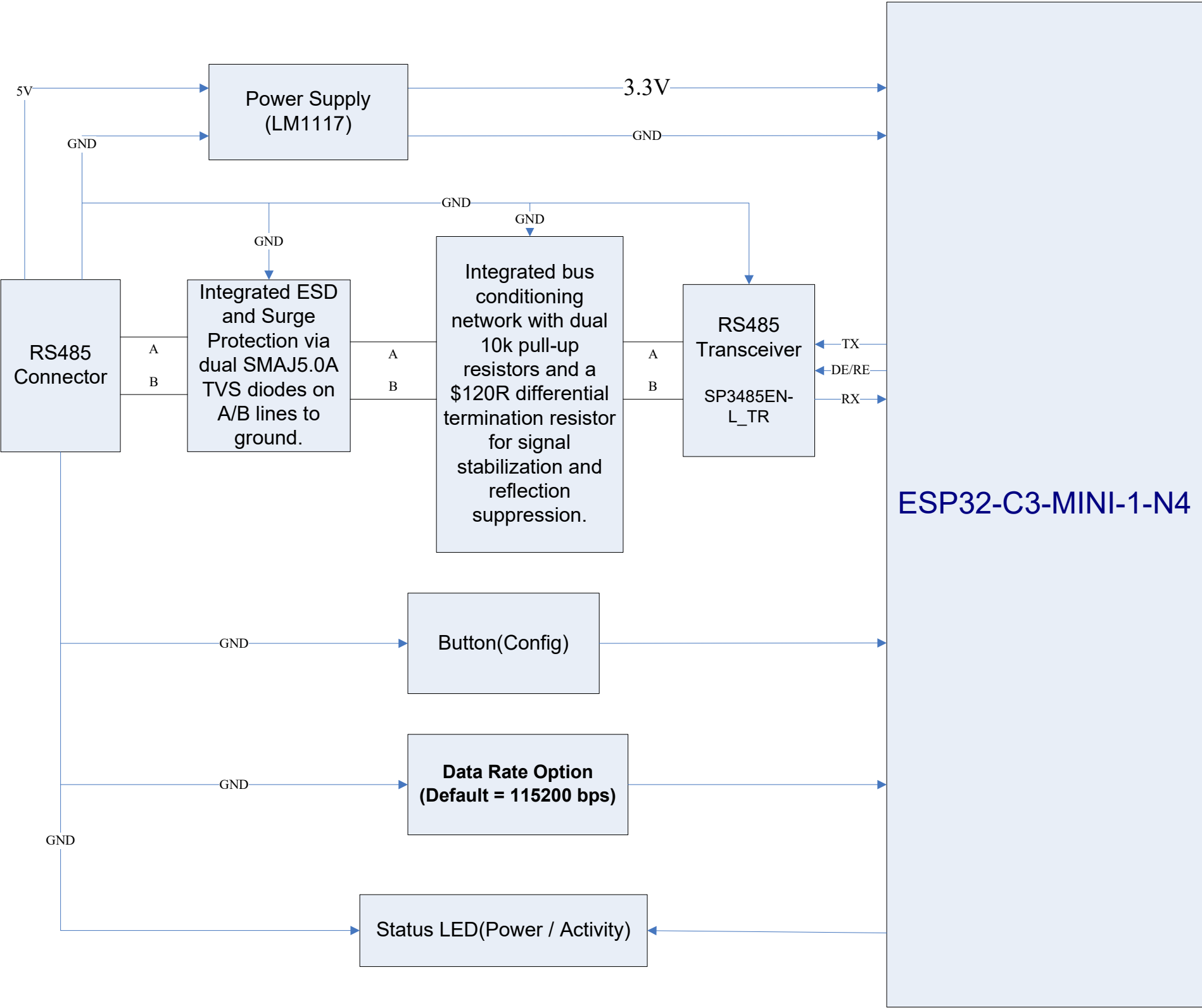


RS485-WiFi-004



The RS485-WiFi-004 is an RS485-to-WiFi bridge module based on the ESP32-C3-MINI-1-N4 microcontroller. External 5V power is regulated to 3.3V by an LM1117 LDO to supply the ESP32 module and the RS485 transceiver. The RS485 interface includes:

- A standard RS485 connector for A, B, and GND lines.
- Integrated ESD and Surge protection via dual SMAJ5.0A TVS diodes on A/B lines to ground per ISO 10605 / IEC 61000-4-2.
- A Bus Conditioning Network consisting of dual 10 kΩ pull-up resistors (R3_4, R4_4) and an integrated 120 Ω differential termination resistor (R6_4) for signal stabilization and reflection suppression.
- A 3.3V RS485 transceiver (SP3485EN-L/TR) interfacing TX/RX/DE signals to the ESP32.

User configuration is supported via a configuration button for data rate selection (Default = 115,200 bps) and status indication through LEDs (Power/Activity). This design ensures robust EMI performance and compliance with FCC Part 15B Class B and ICES-003 Issue 7 requirements.